mbsolver

Two-Level Model

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Vahid Anari and Toktam Rashidi

vahid.anari8@gmail.com

https://github.com/vahid-anari/mbsolver

This document derives the dimensionless two-level Maxwell–Bloch equations solved by `mbsolver.two_level.solve_1d`, starting from the two-level Hamiltonian and proceeding through the rotating-wave and slowly-varying-envelope approximations to the final dimensionless system.

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1 Hamiltonian

Let $|2\rangle$ denote the upper state and $|1\rangle$ the lower state of a two-level system with transition frequency ω_0. The free Hamiltonian is

$$\hat{H}_0 = \frac{\hbar\omega_0}{2}\hat{\sigma}_z, \qquad \hat{\sigma}_z = |2\rangle\langle 2| - |1\rangle\langle 1|, \tag{1}$$

with raising and lowering operators

$$\hat{\sigma}_+ = |2\rangle\langle 1|, \qquad \hat{\sigma}_- = |1\rangle\langle 2|. \tag{2}$$

The electric dipole operator is written in terms of the transition matrix elements $\boldsymbol{d}_{21} = \langle 2|\hat{\boldsymbol{d}}|1\rangle$ and $\boldsymbol{d}_{12} = \boldsymbol{d}_{21}^*$ as

$$\hat{\boldsymbol{d}} = \boldsymbol{d}_{21}\,\hat{\sigma}_+ + \boldsymbol{d}_{12}\,\hat{\sigma}_-. \tag{3}$$

Writing the electric field along polarization unit vector $\hat{\boldsymbol{e}}$ as $\boldsymbol{E} = E\hat{\boldsymbol{e}}$ and defining the scalar transition matrix element

$$d = \boldsymbol{d}_{21}\cdot\hat{\boldsymbol{e}}, \tag{4}$$

the interaction Hamiltonian takes the form

$$\hat{H}_{\text{int}} = -\hat{\boldsymbol{d}}\cdot\boldsymbol{E} = -\big(d\,\hat{\sigma}_+ + d^*\hat{\sigma}_-\big)E. \tag{5}$$

The total Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$.

2 Density-Matrix Equations of Motion

The state of the medium is described by the density operator

$$\hat{\rho} = \sum_{m,n\,\in\,\{1,2\}} \rho_{mn}\,|m\rangle\langle n|, \tag{6}$$

with population inversion

$$w = \rho_{22} - \rho_{11}, \qquad \rho_{21} = \rho_{12}^*. \tag{7}$$

From the Liouville–von Neumann equation $i\hbar\dot{\hat{\rho}} = [\hat{H},\hat{\rho}]$, the matrix element $\langle 1|\,[\hat{H},\hat{\rho}]\,|2\rangle$ gives

$$\begin{aligned} i\hbar\dot{\rho}_{12} &= \langle 1|[\hat{H},\hat{\rho}]|2\rangle \\ &= -\hbar\omega_0\rho_{12} - d^*E\,(\rho_{22}-\rho_{11}), \end{aligned} \tag{8}$$

giving

$$\dot{\rho}_{12} = i\omega_0\rho_{12} + \frac{id^*}{\hbar}\,E\,w. \tag{9}$$

Since $\rho_{21} = \rho_{12}^*$, the complex conjugate gives immediately

$$\dot{\rho}_{21} = -i\omega_0\rho_{21} - \frac{id}{\hbar}\,E\,w. \tag{10}$$

For the inversion, taking $\langle 2|[\hat{H},\hat{\rho}]|2\rangle - \langle 1|[\hat{H},\hat{\rho}]|1\rangle$:

$$i\hbar\dot{w} = 2\big(d^*E\,\rho_{21} - dE\,\rho_{12}\big), \tag{11}$$

so that

$$\dot{w} = \frac{2i}{\hbar}\big(dE\,\rho_{12} - d^*E\,\rho_{21}\big). \tag{12}$$

3 Maxwell Wave Equations

3.1 Maxwell's equations

In a medium with no free charges or currents, Maxwell's equations are

$$\nabla \cdot \mathbf{D} = 0, \tag{13}$$
$$\nabla \cdot \mathbf{B} = 0, \tag{14}$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{15}$$
$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}, \tag{16}$$

with constitutive relations

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \qquad \mathbf{B} = \mu_0 \big(\mathbf{H} + \mathbf{M}\big). \tag{17}$$

3.2 General wave equations

Taking the curl of the curl equations and using the constitutive relations gives the general wave equations for $\mathbf{E}$ and $\mathbf{B}$:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2} + \mu_0 \nabla \times \frac{\partial \mathbf{M}}{\partial t}, \tag{18}$$

$$\nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = -\mu_0 \nabla \times \left(\frac{\partial \mathbf{P}}{\partial t} + \nabla \times \mathbf{M} \right). \tag{19}$$

3.3 Reduction to one dimension

We consider a plane wave propagating along z with fields transverse to z. For the **electric-dipole** case the medium has no magnetization ($\mathbf{M} = 0$), so the curl term in Eq. (18) vanishes and the wave equation reduces to

$$\frac{\partial^2 E}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = \mu_0 \frac{\partial^2 P}{\partial t^2}. \tag{20}$$

The macroscopic polarization is related to the density operator through

$$P = n_0 \operatorname{Tr}(\hat{\rho}\, \hat{\boldsymbol{d}} \cdot \hat{\boldsymbol{e}}) = n_0 \big(\rho_{12}\, d + \rho_{21}\, d^*\big). \tag{21}$$

4 Slowly Varying Envelope and Rotating-Wave Approximations

We apply two standard approximations [2, 3]: the **slowly varying envelope approximation** (SVEA) to the field, and the **rotating-wave approximation** (RWA) to the atom–field coupling.

4.1 Envelope decomposition

The field and coherence ρ_{21} are written as slowly varying envelopes modulated by the carrier $e^{i(kz-\omega t)}$:

$$E(z,t) = \tfrac{1}{2}A(z,t)\,e^{i(kz-\omega t)} + \text{c.c.}, \tag{22}$$

$$\rho_{21}(z,t) = \tfrac{1}{2}R(z,t)\,e^{i(kz-\omega t)}, \tag{23}$$

where $k = \omega/c$ is the carrier wave number, and A and R satisfy

$$\left|\frac{\partial A}{\partial t}\right| \ll \omega|A|, \quad \left|\frac{\partial A}{\partial z}\right| \ll k|A|, \tag{24}$$

and similarly for R. The factor of $\frac{1}{2}$ ensures that A equals the peak amplitude of the field.

4.2 Equations of motion

Substituting the envelope decomposition into Eq. (10), collecting terms at the resonant frequency $e^{i(kz-\omega t)}$, and dropping the fast counter-rotating terms (RWA). we define the detuning

$$\Delta = \omega - \omega_0, \tag{25}$$

the coherence equation becomes

$$\dot{R} = i\Delta R - \frac{id}{\hbar}\,A\,w. \tag{26}$$

Substituting into Eq. (12) and keeping only the two resonant products (one from each factor of the field and coherence):

$$\dot{w} = \frac{i}{2\hbar}\big(dAR^* - d^*A^*R\big). \tag{27}$$

Substituting into the wave equation (20), dropping second-order derivatives of A (SVEA), and using $k = \omega/c$:

$$\frac{\partial A}{\partial z} + \frac{1}{c}\frac{\partial A}{\partial t} = \frac{i\omega\, n_0 d^*}{2\epsilon_0 c}\,R. \tag{28}$$

4.3 Relaxation and pumping

In practice the two-level medium is coupled to its environment, leading to population relaxation on a timescale T_1 and dephasing on a timescale T_2. These are added phenomenologically. A pump term Λ_n drives the inversion away from its equilibrium value w_0, and the coherence relaxes toward its equilibrium value R_0. We also set the detuning $\Delta = 0$, corresponding to exact resonance between the carrier and the atomic transition. The full Maxwell–Bloch system is:

$$\frac{\partial w}{\partial t} = \frac{i}{2\hbar}\big(dAR^* - d^*A^*R\big) - \frac{w - w_0}{T_1} + \Lambda_n, \tag{29}$$

$$\frac{\partial R}{\partial t} = -\frac{id}{\hbar}\,A\,w - \frac{R - R_0}{T_2}, \tag{30}$$

$$\frac{\partial A}{\partial z} + \frac{1}{c}\frac{\partial A}{\partial t} = \frac{i\omega\, n_0 d^*}{2\epsilon_0 c}\,R. \tag{31}$$

5 Retarded-Time Coordinates

We move to a frame co-propagating with the field by introducing the retarded time $\tau = t - z/c$, keeping z as the propagation coordinate. Under this transformation

$$\left.\frac{\partial}{\partial z}\right|_t = \left.\frac{\partial}{\partial z}\right|_\tau - \frac{1}{c}\left.\frac{\partial}{\partial \tau}\right|_z, \qquad \left.\frac{\partial}{\partial t}\right|_z = \left.\frac{\partial}{\partial \tau}\right|_z, \tag{32}$$

so the operator $\partial_z|_t + c^{-1}\partial_t|_z$ reduces to $\partial_z|_\tau$, and the field equation becomes a pure spatial propagation equation. The full Maxwell–Bloch system in retarded-time coordinates is:

$$\frac{\partial w}{\partial \tau} = \frac{i}{2\hbar}\left(dAR^* - d^*A^*R\right) - \frac{w - w_0}{T_1} + \Lambda_n, \tag{33}$$

$$\frac{\partial R}{\partial \tau} = -\frac{id}{\hbar}\,A\,w - \frac{R - R_0}{T_2}, \tag{34}$$

$$\frac{\partial A}{\partial z} = \frac{i\omega\, n_0 d^*}{2\epsilon_0 c}\,R. \tag{35}$$

6 Dimensionless Form

6.1 Scaled variables

The inversion w and coherence R are already dimensionless. We introduce dimensionless forms of the remaining variables by normalising lengths to the sample length L, times to the characteristic scale T_0, and the field amplitude to $\hbar/dT_0$:

$$z' = \frac{z}{L}, \qquad \tau' = \frac{\tau}{T_0}, \qquad T_1' = \frac{T_1}{T_0}, \qquad T_2' = \frac{T_2}{T_0}, \qquad \Lambda_n' = T_0\Lambda_n. \tag{36}$$

To absorb the imaginary unit in the field equation we define the dimensionless field amplitude

$$A' = \frac{A}{A_0}, \tag{37}$$

where the field amplitude scale is

$$A_0 = -i\frac{\hbar}{d\,T_0}. \tag{38}$$

6.2 Cooperation time T_R

Substituting Eqs. (37)–(38) into Eq. (35):

$$\frac{\partial A'}{\partial z'} = -\eta\,R, \tag{39}$$

where

$$\eta = \frac{T_0}{T_R}, \tag{40}$$

and the **cooperation time** is defined as

$$\boxed{T_R = \frac{2\epsilon_0 c\hbar}{\omega\,|d|^2\,n_0\,L}.} \tag{41}$$

6.3 Final dimensionless equations

Substituting Eq. (37) into Eqs. (33)–(35) and expressing all times in units of T_0:

$$\frac{\partial w}{\partial \tau'} = \mathrm{Re}(A' R^*) - \frac{w - w_0}{T_1'} + \Lambda_n', \tag{42}$$

$$\frac{\partial R}{\partial \tau'} = -A'\, w - \frac{R - R_0}{T_2'}, \tag{43}$$

$$\frac{\partial A'}{\partial z'} = -\eta R. \tag{44}$$

Note

No explicit $\hbar$, d, or T_0 appear in Eqs. (42)–(44). All microscopic parameters are absorbed into η and the dimensionless field amplitude A'.

6.4 Symbol table

Symbol	Meaning	Range	Units
$A_0 = -i\hbar/dT_0$	Field amplitude scale	$\mathbb{C}$	(V/m)
$A' = A/A_0$	Dimensionless field envelope	$\mathbb{C}$	—
$R(z', \tau')$	Coherence envelope ($\rho_{21} = \frac{1}{2} R\, e^{i(kz-\omega t)}$)	$\mathbb{C}$	—
$w(z', \tau')$	Population inversion $\rho_{22} - \rho_{11}$	$\mathbb{R}$	—
$z' = z/L$	Normalised propagation distance	[0, 1]	—
$\tau' = \tau/T_0$	Normalised retarded time	$\mathbb{R}^+$	—
T_0	Physical reference time scale	$\mathbb{R}^+$	(s)
$T_1' = T_1/T_0$	Longitudinal relaxation time (may vary with τ')	$\mathbb{R}^+$	—
$T_2' = T_2/T_0$	Transverse dephasing time (may vary with τ')	$\mathbb{R}^+$	—
T_R	Cooperation time (Eq. 41)	$\mathbb{R}^+$	(s)
$\eta = T_0/T_R$	Dimensionless coupling parameter	$\mathbb{R}^+$	—
$\Lambda_n' = T_0 \Lambda_n$	Dimensionless pump rate	$\mathbb{R}$	—
w_0	Equilibrium inversion	$\mathbb{R}$	—

7 Initial and Boundary Conditions

7.1 General case

The Maxwell–Bloch system (42)–(44) requires:

- an **initial condition** for R at $\tau' = 0$ for all z',
- a **boundary condition** for A' at $z' = 0$ for all τ'.

The inversion w is always real since ρ_{11} and ρ_{22} are real probabilities. In the general case R and A' are complex.

7.2 Real-variable reduction

We now show that under a natural choice of initial and boundary conditions, R and A' remain real for all τ' and z', leading to a significant simplification.

Suppose the initial condition for the coherence satisfies

$$R(z', 0) \in \mathbb{R}, \tag{45}$$

and the boundary field is real:

$$A'(0, \tau') \in \mathbb{R}. \tag{46}$$

Note that A' real corresponds to A purely imaginary, since $A' = A/A_0$ with $A_0 = -i\hbar/dT_0$.

Inspecting the equations: if w, R, and A' are all real at some τ', then the right-hand sides of Eqs. (42)–(44) are all real, so the variables remain real for all subsequent times and positions. Under these conditions $\mathrm{Re}(A'R^*) = A'R$, and the Maxwell–Bloch system reduces to:

$$\frac{\partial w}{\partial \tau'} = A'R - \frac{w - w_0}{T_1'} + \Lambda_n', \tag{47}$$

$$\frac{\partial R}{\partial \tau'} = -A'\, w - \frac{R - R_0}{T_2'}, \tag{48}$$

$$\frac{\partial A'}{\partial z'} = -\eta R. \tag{49}$$

Note

Equations (47)–(49) are entirely real and require no complex arithmetic. This is the system solved by `mbsolver.two_level.solve_1d`. The equations hold unchanged when the relaxation times depend on time, $T_1' = T_1'(\tau')$ and $T_2' = T_2'(\tau')$. Section 11 lists the solver argument for each symbol.

8 Recovering the Physical Fields

We define the field and polarization amplitude scales

$$E_0 = \frac{\hbar}{|d|\, T_0}, \qquad P_0 = n_0\, |d|. \tag{50}$$

Given the dimensionless solution $A'(z', \tau')$, the instantaneous electric field and macroscopic polarization are recovered as

$$E(z, t) = E_0\, \mathrm{Re}\big[-iA'(z', \tau')\, e^{i(kz - \omega t)}\big]\,, \tag{51}$$

$$P(z, t) = P_0\, \mathrm{Re}\big[R(z', \tau')\, e^{i(kz - \omega t)}\big]\,. \tag{52}$$

Since the carrier $e^{i(kz-\omega t)}$ oscillates much faster than the envelopes A' and R, physical detectors respond only to the slowly varying amplitudes. The observable intensity is therefore proportional to $|A'|^2$.

In the real case (Subsection 7.2), and taking d real, A' and R are real, and the fields reduce to

$$E(z,t) = E_0\, A'(z',\tau')\sin(kz-\omega t), \tag{53}$$

$$P(z,t) = P_0\, R(z',\tau')\cos(kz-\omega t). \tag{54}$$

9 Magnetic-Dipole Case

All preceding sections were derived for the electric-dipole case. The magnetic-dipole (maser) case follows from the substitutions

$$E \;\to\; cB, \qquad d \;\to\; \frac{m}{c}, \qquad P \;\to\; \frac{M}{c}, \tag{55}$$

where $m = \boldsymbol{m}_{21}\cdot\hat{\boldsymbol{b}}$ is the magnetic transition dipole matrix element and $\hat{\boldsymbol{b}}$ is the magnetic field polarization unit vector. Under these replacements the coupling $dE = (m/c)(cB) = mB$ is preserved, and the dimensionless equations (47)–(49) hold unchanged in form.

The table below summarises the correspondence between the two cases at each stage of the derivation.

Quantity	Electric	Magnetic
Microscopic quantities		
Interaction Hamiltonian	$\hat{H}_{\text{int}} = -\hat{\boldsymbol{d}}\cdot\boldsymbol{E}$	$\hat{H}_{\text{int}} = -\hat{\boldsymbol{m}}\cdot\boldsymbol{B}$
Coupling constant	$d = \boldsymbol{d}_{21}\cdot\hat{\boldsymbol{e}}$	$m = \boldsymbol{m}_{21}\cdot\hat{\boldsymbol{b}}$
1D wave equation	$\dfrac{\partial^2 E}{\partial z^2} - \dfrac{1}{c^2}\dfrac{\partial^2 E}{\partial t^2} = \mu_0\dfrac{\partial^2 P}{\partial t^2}$	$\dfrac{\partial^2 B}{\partial z^2} - \dfrac{1}{c^2}\dfrac{\partial^2 B}{\partial t^2} = \mu_0\dfrac{\partial^2 M}{\partial z^2}$
Medium response	$P = n_0\big(\rho_{12}\,d + \rho_{21}\,d^*\big)$	$M = n_0\big(\rho_{12}\,m + \rho_{21}\,m^*\big)$
Field envelope	$E = \frac{1}{2}A\,e^{i(kz-\omega t)} + \text{c.c.}$	$B = \frac{1}{2}A\,e^{i(kz-\omega t)} + \text{c.c.}$
Scaling quantities		
Cooperation time	$T_R = \dfrac{2\epsilon_0 c\hbar}{\omega\,\lvert d\rvert^2\,n_0\,L}$	$T_R = \dfrac{2c\hbar}{\omega\mu_0\,\lvert m\rvert^2\,n_0\,L}$
Field amplitude scale	$A_0 = -i\dfrac{\hbar}{d\,T_0}$	$A_0 = -i\dfrac{\hbar}{m\,T_0}$
Field scale	$E_0 = \dfrac{\hbar}{\lvert d\rvert\,T_0}$	$B_0 = \dfrac{\hbar}{\lvert m\rvert\,T_0}$
Medium response scale	$P_0 = n_0\,\lvert d\rvert$	$M_0 = n_0\,\lvert m\rvert$
Intensity scale	$I_0 = \dfrac{\epsilon_0 c E_0^2}{2}$	$I_0 = \dfrac{cB_0^2}{2\mu_0}$
Physical observables (real case)		
Physical field	$E_0\,A'\sin(kz-\omega t)$	$B_0\,A'\sin(kz-\omega t)$
Medium response	$P_0\,R\cos(kz-\omega t)$	$M_0\,R\cos(kz-\omega t)$
Intensity	$I_0\,\lvert A'\rvert^2\sin^2(kz-\omega t)$	

In both cases the cooperation time can be expressed in terms of the single-atom spontaneous emission rate Γ. Using the standard relations

$$\Gamma = \frac{\omega^3 |d|^2}{3\pi\epsilon_0 \hbar c^3} \quad \text{(electric)}, \qquad \Gamma = \frac{\omega^3 |m|^2}{3\pi\epsilon_0 \hbar c^5} \quad \text{(magnetic)}, \tag{56}$$

substituting into the respective expressions for T_R yields the same result in both cases:

$$\boxed{T_R = \frac{2\omega^2}{3\pi c^2 \, n_0 \, L \, \Gamma}.} \tag{57}$$

The cooperation time is therefore inversely proportional to n_0, L, and Γ. For a sufficiently dense and inverted medium, $T_R \ll T_1$ and the system enters the superradiant regime [1, 2]. These equations have been applied to astrophysical masers and superradiance, for example to the 21 cm and OH 1612 MHz lines [4, 5], recurring maser flares [6] and the 6.7 GHz methanol flare in S255IR-NIRS3 [7]. The function `mbsolver.scaling` computes T_R, η and the other scales from ν, Γ, T_0, L and n_0 (Section 11).

Note

Because the dimensionless equations (47)–(49) depend only on η, T_1', T_2', w_0, R_0, Λ_n', and the boundary field $A'(0, \tau')$, the same solver describes both electric-dipole and magnetic-dipole physics.

10 Adiabatic (Rate-Equation) Limit

When T_2' is short compared with the times over which A' and w change, the coherence follows the field and the population without delay. Setting $\partial R/\partial \tau' = 0$ in Eq. (48) gives

$$R = R_0 - T_2' \, A' w, \tag{58}$$

and substituting it into Eqs. (47) and (49) leaves two equations, for w and A' only:

$$\frac{\partial w}{\partial \tau'} = A' R_0 - T_2' A'^2 w - \frac{w - w_0}{T_1'} + \Lambda_n', \tag{59}$$

$$\frac{\partial A'}{\partial z'} = -\eta R_0 + \eta \, T_2' A' w. \tag{60}$$

The term $T_2' A'^2 w$ is stimulated emission, which uses up the inversion, and $\eta T_2' w$ is the gain coefficient of the field: a weak field grows as $e^{\eta T_2' w z'}$ through an inverted medium. These are the rate equations of a maser amplifier.

The limit breaks down where the coherence has no time to follow: during the first few T_2', when the full system still has $R = R_0$ while Eq. (58) applies the full gain at once, and in bursts faster than T_2' (superradiant ringing). Late-time, slowly varying solutions agree with the full system for any T_2'. The limit needs a finite T_2'.

Note

`mbsolver.two_level.solve_1d_adiabatic` solves Eqs. (59)–(60) on the same grids as `solve_1d`, with the same explicit step in z'. Over one time step $\Delta\tau'$ the field, T_1', T_2' and Λ_n' are held fixed, so Eq. (59) is linear, $\partial w/\partial\tau' = a - b\,w$ with $a = A'R_0 + w_0/T_1' + \Lambda_n'$ and $b = T_2'A'^2 + 1/T_1'$, and is advanced exactly:

$$w(\tau' + \Delta\tau') = w + (a - b\,w)\,\frac{1 - e^{-b\Delta\tau'}}{b}.$$

A strong field makes $b\,\Delta\tau'$ large (a stiff equation) where Runge–Kutta would diverge; this step is stable for any $\Delta\tau'$. R follows from Eq. (58) and is returned as `res.R`.

11 In the Code

`mbsolver.two_level.solve_1d` solves Eqs. (47)–(49) with fourth-order Runge–Kutta in τ' and an explicit step in z'. Its arguments and results map onto the symbols as follows (all dimensionless, times in units of T_0). `solve_1d_adiabatic` (Section 10) takes the same arguments and returns the same result, plus R.

Symbol	Argument	Result
τ' grid, z' grid	`t_max`, `nt`, `nz`	`res.time`, `res.z`
η	`eta`	—
$w_0 = \cos\theta_0$, $R_0 = \sin\theta_0$	`theta0`, or `w0` and `R0`	—
T_1', T_2'	`t1`, `t2` (number or `Steps`)	`res.t1`, `res.t2`
Λ_n'	`pump` (`Pulse`, number, list)	`res.pump`
$A'(0,\tau')$	`field_in`	`res.field_in`
$w(z',\tau')$	—	`res.w`
$\lvert A'(z',\tau')\rvert^2$	—	`res.intensity`
$R(z',\tau')$	—	`res.R`

`mbsolver.scaling(nu, gamma, t0, l, n0)` takes $\nu = \omega/2\pi$, Γ, T_0, L and n_0 in SI units and returns $\eta = T_0/T_R$ (Eq. 57), $\theta_0 = 2/\sqrt{N}$ with $N = n_0\lambda L^2$, and the scales E_0, P_0, I_0 (and their magnetic counterparts B_0, M_0).

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